

Solutions to problems from Homework 1-2.

1. Theorem If $A_1, \dots, A_n$ are events then
(8+12p)

$$P(A_1 + A_2 + \dots + A_n) = S_1 - S_2 + \dots + (-1)^{n+1} S_n$$

where

$$S_1 = P(A_1) + \dots + P(A_n)$$

$$S_2 = P(A_1 A_2) + P(A_1 A_3) + \dots + P(A_{n-1} A_n)$$

...

$$S_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} P(A_{i_1} A_{i_2} \dots A_{i_k})$$

...

$$S_n = P(A_1 A_2 \dots A_n).$$

Proof (sketch). If A is an event, let

$$1_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \in \Omega - A \end{cases}$$

the so-called indicator (r.v.) of A .

Facts $1_{\bar{A}} = 1 - 1_A$, $1_{AB} = 1_A \cdot 1_B$, $P(A) = \mathbb{E} 1_A$.

$$\begin{aligned} \text{Then } P(A_1 + \dots + A_n) &= 1 - P(\overline{A_1 + \dots + A_n}) = \\ &= 1 - P(\bar{A}_1 \cdot \bar{A}_2 \dots \bar{A}_n) = 1 - \mathbb{E} 1_{\bar{A}_1} \cdot 1_{\bar{A}_2} \dots 1_{\bar{A}_n} = \\ &= 1 - \mathbb{E} (1 - 1_{A_1})(1 - 1_{A_2}) \dots (1 - 1_{A_n}) = \\ &= \mathbb{E} \left[(1_{A_1} + \dots + 1_{A_n}) - (1_{A_1 A_2} + \dots + 1_{A_{n-1} A_n}) + \dots \right] = \dots = \end{aligned}$$

